

Run Rawk on an example human dataset

This tutorial shows how to use Rawk to analyze the Level 5 moderated z-scores of the human L1000 assay dataset of cellular responses to chemical and genetic perturbagens.

Import python dependencies

```
import rawk as rk
import pathlib
import os
import pandas as pd
import numpy as np
```

Set up output directory

```
out_dir = os.path.join(
    "tutorial_output",
    "example_human_data_analysis")
pathlib.Path(out_dir).mkdir(parents=True, exist_ok=True)
```

Read input data

```
original_data_df = pd.read_csv(
    os.path.join(
        "tutorial_data",
        "human_l1000_level5_zscores.csv"))

mrn_node_df = pd.read_table(
    os.path.join(
        "tutorial_output",
        "construct_recon3d_mrn",
```

```

        "recon3d_unfiltered_nodes.tsv"))

mrn_edge_df = pd.read_table(
    os.path.join(
        "tutorial_output",
        "construct_recon3d_mrn",
        "recon3d_pruned_edges.tsv.gz"),
    low_memory=False)

```

Preprocess input data

```

# Select the first gene column and one or more other
# columns for running rawk
#
# Here, select columns gene and
# LJP005_HA1E_24H_G13____HA1E____withaferin-a____10_0_um____24_h
columns_for_rawk = [
    "gene",
    "LJP005_HA1E_24H_G13____HA1E____withaferin-
a____10_0_um____24_h",
]

df_for_rawk_analysis = original_data_df.loc[:,
columns_for_rawk]

def preprocess_1d_arr(x):
    p = np.e ** rk.qn_transform(
        x, log1p=False, collapse_0s=False,
        center=True)
    return p

pp_node_df, pp_edge_df = rk.get_met_net_dfs(
    mrn_node_df, mrn_edge_df,
    df_for_rawk_analysis,
    # edges were pruned, so no additional filter
    0,
    fill_missing_gene_prop=None,
    transform_gene_prop_func=preprocess_1d_arr)

```

Run Rawk

```
msrk = rk.MultiSampleRawk(  
    pp_node_df.drop(columns=["rxn_name", "equation"]),  
    pp_edge_df,  
    n_jobs=1,  
    n2v_walk_length=20,  
    n2v_num_walks=8000,  
    seed=42,  
)  
  
mrn_n_genes_by_pathway = (  
    mrn_node_df  
    .groupby("pathway")["gene"]  
    .apply(lambda x: len(set(x)))  
    .to_dict()  
)  
  
pathways_for_testing = [  
    p for p, n in mrn_n_genes_by_pathway.items()  
    if n >= 5 and p != "Miscellaneous"]  
  
rk_fdr_df, rk_es_df = msrk.test_num_steps(  
    pw_subset=pathways_for_testing)
```

Plot Rawk results

```
plot_pathway = "CoA synthesis"  
plot_rawk = msrk.rawk_list[0]  
# each Rawk instance has a sample and a background sample  
assert plot_rawk.sample.name ==  
df_for_rawk_analysis.columns[1]  
assert plot_rawk.bg_sample.name == "uniform_background"
```

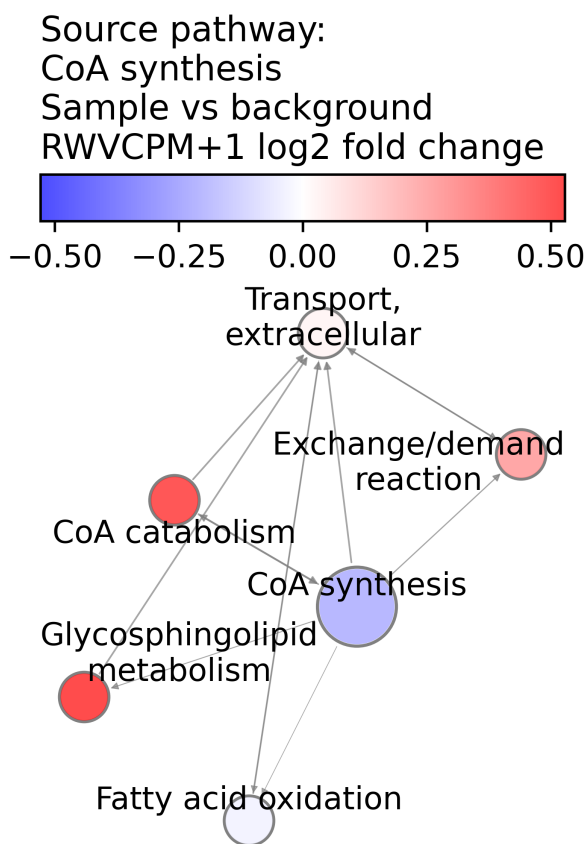
```
rk.plot_pw_neighborhood(  
    plot_rawk,  
    plot_pathway,  
    os.path.join(  

```

```

        out_dir,
        "rawk_pathway_nbr_rwv_cpm_p1_log2fc.png"),
    "rwv_cpm_p1_log2fc",
    node_color_center=0,
    n_cutoff=6, node_alpha=0.7,
    node_color_title=(
        "Sample vs background\n"
        "RWVCPM+1 log2 fold change"
    ),
    title=f"Source pathway:\n{plot_pathway}")

```



```

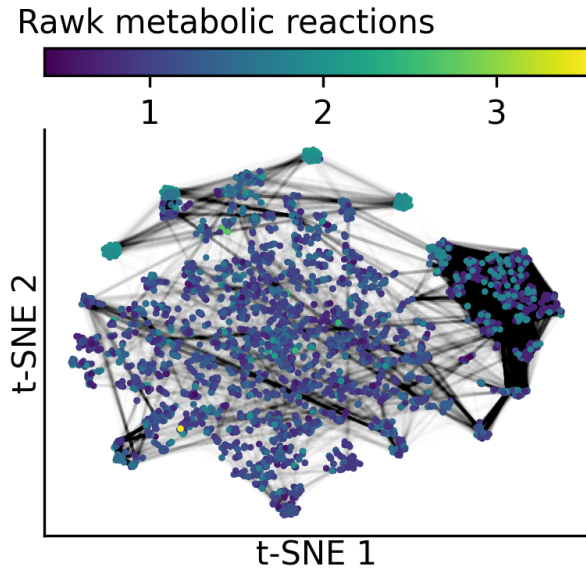
rk.plot_graph(
    plot_rawk.sample.rxn_graph,
    plot_rawk.sample.rxn_pos,
    os.path.join(
        out_dir,
        "rawk_mrn_reaction_tsne.png"),
    draw_edges=True,

```

```

non_s_pw_node_size=4,
title="Rawk metabolic reactions",
dim_name="t-SNE")

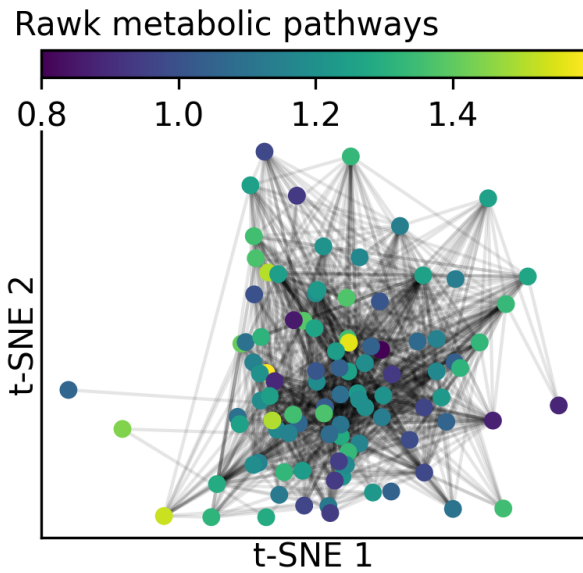
```



```

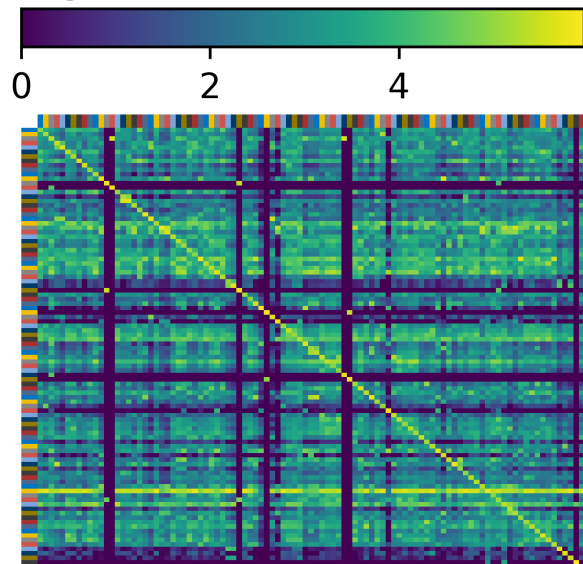
pathway_pos = {
    n: d["pw_pos"]
    for n, d in plot_rawk.sample.pw_graph.nodes.items()
}
rk.plot_graph(
    plot_rawk.sample.pw_graph,
    pathway_pos,
    os.path.join(
        out_dir,
        "rawk_mrn_pathway_tsne.png"),
    node_color_attr="pw_property",
    non_s_pw_node_size=30,
    draw_edges=True,
    title="Rawk metabolic pathways",
    dim_name="t-SNE")

```



```
rk.plot_rawk_sample_mtx(
    plot_rawk.sample,
    os.path.join(
        out_dir, "rawk_test_sample"))
```

Columns: source pathways
Rows: target pathways
 $\log_{10}(\text{RWVCPM} + 1)$



Save Rawk results

```
rk_fdr_df.to_csv(  
    os.path.join(out_dir, "rawk_fdr.tsv"),  
    sep="\t", index=False)  
  
rk_es_df.to_csv(  
    os.path.join(out_dir, "rawk_es.tsv"),  
    sep="\t", index=False)
```

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